Duc V. Le

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(Last updated June 2026.)

Research Interests

Primary Areas: Applied Cryptography, Privacy-Enhancing Technologies, Distributed Systems Security
Specific Focus: Zero-Knowledge Proofs, Ring Signatures, Oblivious RAM, Blockchain Privacy, Payment Channel Security, Consensus Protocol Analysis, Transaction Unlinkability, Privacy-Preserving Financial Systems

Education

Ph.D. in Computer Science 2015–2021
Purdue University

- **Dissertation:** "Efficient Cryptographic Constructions for Resource-Constrained Blockchain Clients"
- **Co-Advisors:** Prof. Aniket P. Kate, Prof. Mikhail J. Atallah
- **Committee:** Prof. Jeremiah Blocki, Prof. Elena Grigorescu
- **Focus:** Applied Cryptography, Blockchain Privacy, Distributed Systems Security

B.S. in Mathematics & Computer Information Systems 2011–2014
University of Mount Union

- **Summa Cum Laude**
- **Double Major:** Mathematics and Computer Information Systems

Professional Experience

Staff Research Engineer March 2026 – Present
Circle Inc.

- **Shaped Circle's post-quantum security strategy**, defining a three-phase migration to quantum-safe cryptography across the Arc blockchain, USDC (30+ chains), smart contracts, validators, and infrastructure using NIST standards (SLH-DSA, hybrid X25519MLKEM768/X-Wing); white paper covered across crypto-industry media.
- **Contributed to core development of the Arc blockchain**, spanning its consensus protocol, privacy layer, and agent stack.
- **Proved impossibility of anti-plutocratic DAO governance**, showing every wallet-balance voting rule (incl. quadratic voting) is asymptotically plutocratic under Sybil splitting, validated across five major DAOs (ENS, Compound, Uniswap, Arbitrum, ZKsync) (under submission).

Staff Research Scientist February 2023 – March 2026
Visa Inc.

- **DARPA P5: Provably Private and Performant Payment Paradigms** Contributed to DARPA RFI response (DARPA-PS-26-23) defining formal privacy notions for scalable international payment systems, including amount confidentiality, k-anonymity, address unlinkability, and sender–receiver unlinkability. Collaborated with RTX BBN Technologies, Purdue University, George Mason University, Amazon, Mysten Labs, Optimex, and Supra Research.
- **Quantum-Safe Blockchain Architecture** Leading quantum threat assessment protecting 300B+ stablecoin ecosystem, producing white paper, blog post, and academic SoK analyzing attack vectors across application, consensus, and network layers. Quantified migration trade-offs. Architecting quantum-safe stablecoin system deployable without underlying blockchain upgrades (On-going).
- **Designed `OptiBridge`, a trustless bridge** that enables cross-chain transfers between Lightning Network payment channels and Ethereum without modifications to either systems (Patent filed, ACNS '26).
- **Data Provenance for LLM Agents** designing a system that provides authenticated data sources for LLM agents (On-going)

- **Contributed to blockchain tokenization platform (VTAP)** enabling banks to issue and manage fiat-backed tokens on public blockchains, providing APIs for minting, burning, and cross-chain interoperability (Patent filed)
- **Led development of Data Tumbling Layer (DTL)** - a composable cryptographic primitive enabling unlinkable data exchange for smart contracts with sub-1.5s transaction initiation, enabling privacy-preserving financial services and voting systems (Solidity, snarkjs, circom) (Patent filed, AsiaCCS ’25)
- **Designed auditable payment channel system** balancing regulatory compliance with transaction privacy for high-throughput off-chain payments, enabling banks to meet regulatory requirements while preserving customer privacy (TypeScript, Circom, Snarkjs) (Patent filed, CBT ’25)
- **Built scalable off-chain auction system** supporting 1000+ bidders with $O(k)$ on-chain complexity for k misbehaving participants, enabling large-scale NFT and digital asset auctions with 95% gas reduction vs existing solutions (Circom, Snarkjs, Solidity) (Patent filed, to appear at NDSS ’26)
- **Optimized Powers-of-Tau ceremony (LitePoT)** achieving 16× gas reduction and supporting up to 2^{15} degree polynomials via fraud-proof mechanism and proof aggregation, enabling cost-effective deployment of zero-knowledge proof systems for blockchain applications (Solidity, TypeScript, Rust) (Patent filed, CCS ’25)

Postdoctoral Researcher October 2021 – January 2023

University of Bern, Postdoc Supervisor: Prof. Christian Cachin

- **Developed formalization of resource-based consensus protocols** creating unified framework for comparing Proof-of-Work, Proof-of-Stake, and Proof-of-Storage systems, enabling security analysis of long-range and nothing-at-stake attacks across different blockchain architectures (OPODIS ’22)
- **Led development of Digital Signatures with Key Extraction (DSKE)** framework enabling automatic private key extraction after threshold signatures, implemented and evaluated both hash-based and polynomial commitment-based constructions demonstrating practical performance, enabling cryptographic deniability and spam prevention (CT-RSA ’25, presented at conference)
- **Led design of Merkle Pyramid Builder approach** achieving 7× reduction in on-chain mixer deposit costs through batched Merkle tree updates, reducing average transaction fees from 1.1M to 300K gas to improve blockchain privacy accessibility (AFT ’23)
- **Developed Privacy-preserving transaction DAG (PDAG)** framework providing unified model for analyzing blockchain privacy, enabling formal comparison of untraceability and unlinkability properties across Monero, Zcash, and mixing protocols (FC ’24)

Research Intern May 2020 – August 2020

Imperial College London

- **Engineered AMR autonomous cryptocurrency mixer** with privacy-preserving reward distribution, supporting 66,000+ deposits per day and anonymity sets exceeding 1,000 users while enabling DeFi yield farming on deposited funds (Solidity, zkSNARKs) (AFT ’21, presented at conference)
- **Co-analyzed sandwich attacks on decentralized exchanges** formalizing front/back-running strategies that can potentially generate 3,400+ USD daily revenue, contributing to mitigation discussions for protecting traders from MEV exploitation (IEEE S&P ’21)

Research Intern May 2019 – August 2019

TU Vienna

- **Co-designed Dual Linkable Spontaneous Anonymous Group (DLSAG) signature scheme** enabling non-interactive refund transactions and payment channels in Monero while maintaining privacy guarantees with 7% faster performance than existing LSAG (FC ’20)
- **Implemented C++ cryptographic library** using libsodium and ed25519 curve, demonstrating practical feasibility for Monero integration with comprehensive performance benchmarks
- **Contributed to Monero community adoption discussions** with open-sourced prototype actively considered for integration into privacy-focused cryptocurrency protocols

Technical Skills

Languages: Java, C++, Python, JavaScript, Solidity, Rust
Domain Expertise: Applied Cryptography, Distributed Systems, Privacy Enhancing Technologies (PETs), Blockchain Privacy, Zero-Knowledge Proofs, Oblivious Access Mechanisms (ORAM), Ring Signatures, Payment Channels, Transaction Unlinkability, Privacy Analysis.

Selected Open-Source Projects

T^3**: Privacy-Preserving Blockchain Client** *GitHub*

- **Built high-performance distributed system** handling 1000+ concurrent requests for private blockchain data access using Intel SGX and ORAM protocols (PETS '20)

DLSAG: Privacy-Preserving Payment Channels *GitHub*

- **Developed production-ready C++ implementation** of dual linkable anonymous group signatures for Monero payment channels with comprehensive testing suite (FC '20)

Professional Service

Program Committee Member

- **Top-Tier Security Conferences:**
  - IEEE Symposium on Security and Privacy (S&P): 2025 (PC), 2022 (Poster Jury)
  - ACM Computer and Communications Security (CCS): 2023–2025 (PC), 2021 (Poster Jury)
  - Network and Distributed System Security Symposium (NDSS): 2023 (Student Support Committee)
- **Privacy and Cryptography Conferences:**
  - Privacy Enhancing Technologies Symposium (PETS): 2024–2025 (PC)
  - Financial Cryptography and Data Security (FC): 2023–2024 (PC)
  - Applied Cryptography and Network Security (ACNS): 2025 (PC)
  - ACM Asia Conference on Computer and Communications Security (ASIACCS): 2025 (PC)
- **Specialized Venues:**
  - IEEE Security and Privacy on the Blockchain (S&B): 2021 (PC)
  - ACM Symposium on Access Control Models and Technologies (SACMAT): 2022–2023 (PC)
  - ConsensusDays: 2022–2023 (PC)

Journal Reviewer

- ACM Transactions on Privacy and Security (TOPS)
- IEEE Transactions on Dependable and Secure Computing (TDSC)

External Reviewer

- International Association for Cryptologic Research (IACR) Crypto: 2025
- IEEE European Symposium on Security and Privacy (EuroS&P): 2021
- Privacy Enhancing Technologies Symposium (PETS): 2023
- The Science of Blockchain Conference (SBC): 2023

Teaching and Mentoring

Student Supervision

- Lucien K. L. Ng:
  - Power of Tau ceremony optimization (Resulted in publication at CCS '25)
  - A Plug-and-Play Long-Range Defense System for Proof-of-Stake Blockchains, resulted in publication at ESORICS '24
- François-Xavier Wicht: Master thesis on Blockchain Privacy Notions, resulted in publication at FC '24
- Marko Cirkovic: Bachelor thesis on Cost-Effective Onchain Mixers, resulted in publication at AFT '23

- Lizzy Tengana Hurtado: Research on Oblivious Database for Blockchain Clients, resulted in publication at PETS '20

Teaching Experience

- Teaching Assistant, Purdue University (CS182 (Discrete Math), CS381 (Algorithms and Data Structures), CS555 (Cryptography), CS528 (Network Security))

Invited Talks

- CT-RSA 2025: "Digital Signature with Key Extraction"
- IC3 Winter Retreat 2023: "Modelling Resources in Permissionless Longest-chain Total-order Broadcast"
- OPODIS 2022: "Modelling Resources in Permissionless Longest-chain Total-order Broadcast"
- Security and Privacy Group, TU Vienna: "T^3: Scaling Oblivious Accesses to Large-scale blockchain"
- Chinese University of Hong Kong: "ORAM and T^3"
- Brave Research Team: "Autonomous Coin Mixer"
- Bitcoin 2019 (Scaling Bitcoin): "T^3: Scaling Oblivious Accesses to Large-scale blockchain"
- Esorics 2019: "Flexible Digital Signature"

Publications

(alphabetical ordering. † indicates student mentee. ‡ indicates project leadership.)*

Whitepapers & Technical Reports

1. Circle's Post-Quantum Security Roadmap: Securing Blockchains, Smart Contracts, and Digital Assets for the Quantum Era [link]
   Mira Belenkiy, **Duc V. Le**, Gordon Liao, Vipin Singh Sehrawat, Dragos Rotaru, Sergey Gorbunov, Milap Sheth, Jay Logelin, Anthony De Abreu, Dan Boneh
   Circle Technical Whitepaper, 2026
2. Preparing for the Quantum Era: Assessing Quantum Computing Risks to Blockchain Security [link]
   Duc V. Le, Panos Chatzigiannis, Navid Alamati, Sunpreet Singh Arora, Anderson Nascimento, Timothy Conard, Noah Levine, Adam Clark, Cuy Sheffield
   Visa Technical Whitepaper, 2026

Peer-Reviewed Conference Publications

1. OptiBridge: A Trustless, Cost-Efficient Bridge Between the Lightning Network and Ethereum
   Pedro Moreno-Sanchez, **Duc V. Le**, Mohsen Minaei
   Applied Cryptography and Network Security (ACNS), 2026
2. Scalable Off-Chain Auction System‡
   Mohsen Minaei, Ranjit Kumaresan, Andrew Beams, Pedro Moreno-Sanchez, Yibin Yang, Srinivasan Raghuraman, Mahdi Zamani, **Duc V. Le**
   Network and Distributed System Security Symposium (NDSS), 2026
3. AUPC: Auditable Unlinkable Payment Channel‡
   Pedro Moreno-Sanchez, Mohsen Minaei, Srinivasan Raghuraman, Panagiotis Chatzigiannis, **Duc V. Le**
   International Workshop on Cryptocurrencies and Blockchain Technology CBT 2025
4. LitePoT: Practical Powers-of-Tau Setup Ceremony‡
   Lucien K.L Ng†, Mohsen Minaei, Adithya Bhat, Pedro Moreno-Sanchez, Panagiotis Chatzigiannis, **Duc V. Le**
   ACM Conference on Computer and Communications Security (CCS) '25
5. DTL: Data Tumbling Layer - A Composable Unlinkability Primitive for Smart Contracts‡
   Mohsen Minaei, Pedro Moreno Sanchez, Zhiyong Fang, Srinivasan Raghuraman, Navid Alamati, Panagiotis

Chatzigiannis, Ranjit Kumaresan, **Duc V. Le**
ACM ASIA Conference on Computer and Communications Security (AsiaCCS), 2025

6. DSKE: Digital Signature With Key Extraction[‡]
   Orestis Alpos, Zhipeng Wang, Alireza Kasouvi, Sze Yiu Chau, **Duc V. Le**, Christian Cachin
   RSA Conference Cryptographers' Track (CT-RSA), 2025
7. BlindPerm: Efficient MEV Mitigation with an Encrypted Mempool and Permutation
   Alireza Kavousi, **Duc V. Le**, Philipp Jovanovic, George Danezis
   International Conference on Principles of Distributed Systems (OPODIS), 2025
8. A Plug-and-Play Long-Range Defense System for Proof-of-Stake Blockchains
   Lucien K.L. Ng[†], Panagiotis Chatzigiannis, **Duc V. Le**, Mohsen Minaei, Ranjit Kumaresan, Mahdi Zamani
   European Symposium on Research in Computer Security (ESORICS), 2024
9. Programmable Payment Channels
   Yibin Yang, Mohsen Minaei, Srinivasan Raghuraman, Ranjit Kumaresan, **Duc V. Le**, Mahdi Zamani
   Applied Cryptography and Network Security (ACNS), 2024
10. Towards Precise Reporting of Cryptographic Misuses
    Yikang Chen, Yibo Liu, **Duc V. Le**, Sze Yiu Chau
    Network and Distributed System Security Symposium (NDSS), 2024
11. A Transaction-Level Model for Blockchain Privacy[‡]
    François-Xavier Wicht[†], Zhipeng Wang, **Duc V. Le**, Christian Cachin
    Financial Cryptography and Data Security (FC), 2024
12. Pay Less for Your Privacy: Towards Cost-Effective On-Chain Mixers[‡]
    Zhipeng Wang, Marko Cirkovic[†], **Duc V. Le**, William Knottenbelt, Christian Cachin
    Advances in Financial Technologies (AFT), 2023
13. Modelling Resources in Permissionless Longest-chain Total-order Broadcast[*,‡]
    Sarah Azouvi, Christian Cachin, **Duc V. Le**, Marko Vukolic, Luca Zanolini
    International Conference on Principles of Distributed Systems (OPODIS), 2022
14. Autonomous Coin Mixer with Privacy Preserving Reward Distribution[‡]
    Duc V. Le, Arthur Gervais
    Advances in Financial Technologies (AFT), 2021
15. High-Frequency Trading on Decentralized On-Chain Exchanges
    Liyi Zhou, Kaihua Qin, Christof Ferreira Torres, **Duc V. Le**, Arthur Gervais
    IEEE Symposium on Security and Privacy (S&P), 2021
16. T^3: Scaling Oblivious Accesses to Large-Scale Blockchain[‡]
    Duc V. Le, Lizzy Tengana Hurtado[†], Adil Ahmad, Mohsen Minaei, Byoungyoung Lee, Aniket Kate
    Privacy Enhancing Technologies Symposium (PETS), 2020
17. DLSAG: Dual Linkable Spontaneous Anonymous Group Signatures
    Pedro Moreno-Sanchez, Arthur Blue, **Duc V. Le**, Sarang Noether, Brandon Goodell, Aniket Kate
    Financial Cryptography and Data Security (FC), 2020
18. Flexible Digital Signatures[‡]
    Duc V. Le, Mahimna Kelkar, Aniket Kate
    European Symposium on Research in Computer Security (ESORICS), 2019

Under Submission / In Preparation

1. Concave is the New Linear: The Impossibility of Anti-Plutocratic DAO Governance
   Austin Bennett, Preston Vander Vos, **Duc V. Le**, Mira Belenkiy
2. FPS: Flexible Payment System
   Adithya Bhat, Srinivasan Raghuraman, Panagiotis Chatzigiannis, **Duc V. Le**, Mohsen Minaei

Honors and Awards

Purdue University

- Travel Grant: Scaling Bitcoin 2019, ESORICS 2019
- Summer Research Grant 2017, 2019

University of Mount Union

- The Ullman Mathematics Prize 2015
- The Alumni Computer Science and Information Systems Senior Prize 2014
- Nordson Scholarship Recipient 2014
- Faculty/Staff Junior Academic Prize 2013
- The Wilbur & Burdekka Stuckey Carl Mathematics Prize 2012